Yarroju Rithvik

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Education

Visvesvaraya National Institute of Technology (VNIT), Nagpur — Aug 2024 – Jul 2026
M.Tech in Applied Artificial Intelligence (Online) — CGPA: 7.63 — *Nagpur, India*

Indian Institute of Space Science and Technology (IIST) — Aug 2024 – Jul 2026
M.Tech in Aerodynamics and Flight Mechanics — CGPA: 8.00 — *Thiruvananthapuram, India*

Indian Institute of Petroleum and Energy (IIPE) — Aug 2016 – Jul 2020
B.Tech in Chemical Engineering — CGPA: 8.11 — *Visakhapatnam, India*

Experience

Finlatics — Feb 2026 – May 2026
Data Science / Machine Learning Intern — *Remote*

- Built machine-learning workflows for banking classification, Facebook Live Sellers clustering, and sales prediction using Python, Pandas, scikit-learn, Matplotlib, and Seaborn.
- Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, clustering diagnostics, and regression error metrics.

Gameplex Technologies — Sep 2021 – Mar 2022
Machine Learning Intern — *Remote*

- Prototyped recommendation and chat-assistant models with feature engineering and Python-based model experiments.
- Built and maintained preprocessing pipelines used for model training and inference workflows.

Projects

Industrial Defect Vision Lab | *Python, Computer Vision, FastAPI, Docker, Streamlit* — GitHub

- Built a reliability-first defect detection pipeline with anomaly localization, robustness testing, and human-review routing.
- Served the workflow through FastAPI/Docker and developed a Streamlit dashboard for inspection review and visualization.

Retail Pricing Optimization | *Python, CatBoost, Pandas, scikit-learn, Streamlit* — GitHub

- Developed a profit-aware pricing platform using Dunnhumby data, CatBoost demand forecasting, elasticity analysis, and promotion intelligence.
- Created a Streamlit dashboard for pricing insights, demand behavior, and decision-support visualization.

Aerostat AI Stabilization | *Python, Sensor Fusion, Motion Prediction, Control* — GitHub

- Built a simulation-based AI and control pipeline for stabilizing a tethered aerostat under wind-disturbance scenarios.
- Integrated sensor-fusion and motion-prediction components for disturbance-aware state estimation and evaluation.

EuroSAT Remote Sensing KAN | *Python, Deep Learning, KANs, Remote Sensing* — GitHub

- Compared Kolmogorov-Arnold Network classifier heads with MLP heads for EuroSAT land-use classification.
- Evaluated satellite image classification with accuracy, F1-score, recall, MCC, ROC-AUC, and loss-based diagnostics.

Technical Skills

Languages: Python, SQL, C/C++, MATLAB
ML / Data Science: NumPy, Pandas, scikit-learn, CatBoost, Regression, Classification, Clustering, Model Evaluation
Deep Learning / Vision: PyTorch, TensorFlow, Keras, OpenCV, CNNs, Image Classification, Anomaly Detection, Remote Sensing
MLOps / Tools: Git, Docker, FastAPI, Streamlit, MLflow, Linux/Unix, Experiment Tracking, Model Deployment
Research Areas: Kolmogorov-Arnold Networks, Interpretable AI, Reinforcement Learning, Physics-Informed ML, Engineering AI